

## 12 · Cost Management Plan

The single most important financial fact in this whole handoff: **today the platform costs almost nothing because AWS credits cover it — in the Institute’s own account, that becomes real money, and almost all of it is AI inference.** This document explains where the cost comes from, what drives it up or down, and the concrete levers and day-one steps to keep it under control.

Figures here are the **real billing numbers** from the audit (01 · Project Audit §3, AWS Cost Explorer, Jan–Aug 2026); this document turns them into a forward plan.

Companion documents: 01 · Project Audit (the raw cost report), 03 · Engine Comparison (per-engine cost), 09 · Product Roadmap (cheap-infra wins). Glossary in the README.

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### Part A · The headline

- **Gross AWS usage, Jan–Aug 2026:  $\approx$  \$371.** Net out-of-pocket: **effectively zero** (a few cents) — because the shared/university account’s bill is covered by **AWS credits**.
- **The Institute’s own account will not have those credits.** That  $\sim$ \$371 of usage becomes real money.
- **$\approx$  98% of the cost is AI (Amazon Bedrock LLM) inference** — reading documents and generating briefings. Everything else — servers, database, storage, delivery — totaled  **$\sim$ \$7 over seven months ( $\sim$ \$1/month)** and fits inside AWS’s always-free allowances at this scale.

**The one-sentence model:** *cost scales with how much AI extraction and generation you run — not with uptime.* An idle platform costs about a dollar a month; a platform re-processing large corpora costs real money.

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### Part B · Where the money goes

From the real billing data (Jan–Aug 2026, gross):

| Category                                                              | Gross                         | Share                       | What it is                                                  |
|-----------------------------------------------------------------------|-------------------------------|-----------------------------|-------------------------------------------------------------|
| <b>Bedrock LLM inference</b> (Claude Haiku/Sonnet, Llama, Titan, BDA) | <b><math>\sim</math>\$364</b> | <b><math>\sim</math>98%</b> | Document extraction, labeling, eval loops, brief generation |
| Amazon Textract                                                       | \$0.63                        | <1%                         | OCR (low — was SCP-blocked in the sandbox)                  |
| EC2/VPC (one-off)                                                     | $\sim$ \$4.37                 | $\sim$ 1%                   | A corpus/embedding build box                                |

| Category                                                        | Gross      | Share | What it is                    |
|-----------------------------------------------------------------|------------|-------|-------------------------------|
| DynamoDB / S3 /<br>Config /<br>CloudWatch /<br>CloudFront / SQS | <\$3 total | ~1%   | The serverless infrastructure |

**And it is bursty.** Of the ~\$371, **\$366 was a single month (July)** — the capstone crunch of re-running extraction over corpora and eval loops. Months with no development ran **~\$0.14–\$4/month**. This is *development* volume, not steady-state serving.

## Part C · What it will cost the Institute (planning anchors)

Grounded in the measured usage:

| Scenario                                                   | Cost                               | Notes                                                                                                  |
|------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Idle</b> (running, occasional demo, no bulk extraction) | <b>~\$1–3/month</b>                | Non-AI infra rate; fits free tiers; Bedrock $\approx$ \$0 when not extracting                          |
| <b>Cold storage</b> (compute torn down, data kept)         | <b>~\$1/month</b>                  | S3 + a database export                                                                                 |
| <b>Torn down entirely</b>                                  | <b>\$0</b>                         | <b>sst remove</b> (note: production is <b>retain</b> , so its buckets survive and need manual cleanup) |
| <b>Active development / bulk re-extraction</b>             | <b>~\$50–350/month</b>             | The measured heavy-dev range                                                                           |
| <b>Steady serving to end users</b> (no bulk re-extraction) | <b>a fraction of the dev range</b> | Variable cost is dominated by extraction volume                                                        |

The variable cost is **entirely LLM inference**, proportional to how many documents are extracted and how many briefings are generated.

## Part D · The levers (how to control it)

**Biggest lever — control extraction/generation volume:** - **Extract on demand, not in bulk.** The costly pattern is re-running extraction across whole corpora. Steady-state serving of already-extracted data is cheap. - **Choose the cheaper engine where appropriate.** Per the [engine comparison \(03\)](#), on the same 8-document test the engines cost **~\$2.13 (text-layer) vs. ~\$3.44 (Texttract-LAYOUT) vs. ~\$7.92 (BDA)** — BDA is  $\sim 3.7\times$  the cheapest. Text-layer is the residency-friendly, lowest-cost option for born-digital PDFs; Texttract-LAYOUT is the recommended default for quality. Matching engine to need is a direct cost lever. - **Cache / avoid re-work.** Don't re-extract unchanged documents; the platform already avoids overwriting a RAP once progress is recorded.

**Cheap infrastructure wins** (small config changes, from [09 · Roadmap](#)): - **Run Lambdas on arm64** (Graviton) — roughly **20% cheaper** compute. - **Add S3 lifecycle rules** (expire old uploads/exports) and **DynamoDB TTL** on staging rows. - **Reduce CloudWatch log retention** from the 30-day default where not needed.

These infra levers matter less in absolute dollars (~\$1/month base) but are free wins and good hygiene as traffic grows.

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## Part E · Day-one actions (do these when you take the account)

1. **Set an AWS Budget alarm immediately** — e.g. alert at \$25, \$50, \$100/month. This is the single most important safeguard; it turns a runaway extraction job into an email, not a surprise invoice.
  2. **Confirm whether any credits apply** to the Institute’s account (most won’t have the sandbox’s) — and plan for the real rate if not.
  3. **Watch the first real extraction run** and note its cost, so you have a per-document anchor for your own usage.
  4. **Turn on Cost Explorer** and check it monthly (there is **no in-app cost dashboard** — cost monitoring is done in the AWS console; see [08 · Content Stewardship §H](#)).
  5. **Right-size before scaling** — apply the arm64 / lifecycle / TTL wins (Part D) before any large corpus run.
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## Part F · What to watch as the platform grows

- **Bulk corpus ingestion is the cost event to plan for.** If the Institute decides to extract a large library of real RAPs (see [09 · Roadmap](#)), model that as a **one-time project cost**, estimate it from a small pilot run first, and set a budget for it — don’t run it open-ended.
  - **Briefing/Q&A generation** also consumes inference; heavy use of the legal-cases assistant adds variable cost.
  - **SES email** (the digest) is negligible at this scale but requires production access (see [09](#)).
  - **Steady-state serving stays cheap** — the serverless stack scales to near-zero when idle, which is the platform’s cost advantage.
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*This plan is anchored in real billing data (net  $\approx$  \$0 in the credit-covered sandbox; ~\$371 gross, ~98% Bedrock inference, with \$366 of it a single development month). The takeaway for the Institute: budget for **AI inference proportional to how much you extract/generate**, protect the account with a Budget alarm on day one, and treat any bulk re-extraction as a planned, estimated project rather than a background activity.*